31. How many triangles can be formed by joining the vertices of a hexagon?

32. In how many ways can 3 vowels and 2 consonants be chosen from {E,Q,U,A,T,I,O,N}?

33. Different the following with respect to x:
(i) $e^x(x^2-5)$ (ii) $\frac{x^2-1}{x+7}$

34. Find $\int (3x^3-2x^2+6x-7)\,dx$

35. Find $\int (\frac{2}{x}+\frac{2}{x^2}-e^x+3)\,dx$

36. Find $\int_0^2 \frac{3}{4}(2-x)\,dx$

37. Find $\int_{-1}^1 2x\,dx$

38. Evaluate $\int_0^2 \int_0^1 [2x+3y]\,dydx$

V. Calculate the following :

39. A class contains 12 boys and 10 girls. From the class, 10 students are to be chosen for a competition under the condition that at least 4 boys and at least 4 girls must be represented. In how many ways can the selection are made

40. What is the number of ways of choosing 4cards from a pack of 52 playing cards? In how many of these (i) 3 cards are of the same suit (ii) 4 cards are belong to different suits (iii) two are red cards and two are black cards

41. There are 3 questions in the first section, 3 questions in the second section and 2 questions in the third section in a question paper of an exam. The student has to answer any 5 questions, choosing atleast one from each section. In how many ways can the student answer the exam?

42. Differentiate the following with respect to x:
(i) $(x^3-4x^2+2x)(e^x+5)$ (ii) $\frac{x^2-7x}{x^2+8}$

43. Integrate the following with respect to x: (i) $\frac{x^2-5x+6}{x}$ (ii) $2\sqrt{x}+\frac{3}{x^4}$

44. Evaluate: (i) $\int_0^2 \frac{3}{4}x(2-x)\,dx$ (ii) $\int_1^2 \frac{12}{x}dx$